PROBLEM 12 | Problems T2_CrystStruct.pdf

Crystalline Structures & Miller Indices

Drawing Direction Vectors in Cubic Cells

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> *12. Draw direction vectors in unit cells for the following cubic directions: a)* $[1\,1\,2]$ *b)* $[\bar{3}\,3\,1]$ *c)* $[2\,1\,2]$ *d)* $[\bar{1}\,0\,1]$ *e)* $[3\,2\,1]$ *f)* $[1\,2\,2]$ *g)* $[\bar{1}\,2\,3]$ *h)* $[0\,\bar{2}\,1]$ *i)* $[2\,\bar{3}\,3]$ *j)* $[1\,2\,\bar{1}]$ *k)* $[2\,2\,3]$ *l)* $[\bar{1}\,0\,3]$

PHASE 1: HYPOTHESIS & PARAMETERS

Geometric Context:

Twelve crystallographic vectors are drawn inside a standard cubic unit cell of normalised dimensions $1 \times 1 \times 1$.

Normalisation and Origin Translation Rules [Session 3 Slides 37-40]:

1. The components of the direction vector must be reduced by dividing by the component of largest absolute value ($u_{\max}$) so that the vector fits strictly inside the volume of a single unit cell:

$$\vec{r} = \left(\frac{u}{u_{\max}}, \frac{v}{u_{\max}}, \frac{w}{u_{\max}} \right)$$

2. If any component is negative (overbar $\bar{u}$), the **origin of the vector must be translated** to the position $+1$ along that axis so that the vector points towards the interior of the cell.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

Citing the systematic procedure [Session 3 Slide 40]:

1. Identify whether there are negative indices and fix the local origin:
2. $\bar{u} \implies$ origin at $x = 1$.
3. $\bar{v} \implies$ origin at $y = 1$.
4. $\bar{w} \implies$ origin at $z = 1$.
5. Draw the vector from the origin (x_0, y_0, z_0) to the end point $(x_0 + \Delta x, y_0 + \Delta y, z_0 + \Delta z)$.
6. Join both points with an oriented arrow.

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

1. **Parallelism:** If we multiply a vector by a positive scalar constant, the direction does not change: $[1\,1\,2]$ and $[2\,2\,4]$ represent exactly the **same crystallographic direction**.
2. **Equivalence of $\langle u\,v\,w \rangle$ Families:** Directions such as $[1\,2\,2]$ and $[2\,1\,2]$ are symmetric members of the $\langle 221 \rangle$ family in the cubic lattice, which implies that they have identical translational repeat length and the same physical properties of conductivity or elastic modulus.